

FEASIBILITY STUDY

FloodSense: An Augmented Reality Mobile Application with Machine Learning-Based Flood Susceptibility Mapping for Bacoor City

Presented to
Department of Computer Studies
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OVERVIEW

Flooding remains one of the most persistent and consequential natural hazards affecting Bacoor City, Cavite. Located in a low-lying coastal area within Cavite province, Bacoor City is regularly exposed to flooding driven by monsoon rainfall, storm surges, and inadequate drainage infrastructure. As supported by the findings of the primary data gathering phase of this study, the impact of flooding on residents is not only physical but also informational — residents consistently lack timely, accurate, and location-specific flood susceptibility information that would allow them to make informed decisions before and during flood events.

A structured survey administered to 50 residents across 12 flood-prone barangays in Bacoor City revealed significant gaps in the current state of flood information dissemination. Of all respondents, 92% reported experiencing flooding in their area, with 54% experiencing it almost every rainy season and 58% reporting severe flooding. Despite the frequency and severity of flooding, only 20% of respondents considered themselves well-informed about flood-prone areas in Bacoor City — with the remaining 80% falling under "Somewhat Aware," "Not Sure," or "Not Aware at All." The three Likert-scale items on flood information quality (Q9–Q11) yielded mean scores of 2.62, 2.70, and 2.72 respectively, indicating near-dissatisfied perceptions of current information quality, timeliness, and clarity. Most critically, 64% of respondents — 32 out of 50 — reported having made a wrong decision due to insufficient flood information, including taking flooded roads, failing to evacuate in time, and misjudging flood conditions.

Thematic analysis of open-ended survey responses further confirmed the nature of this gap. The two most prevalent themes from Q12 — "Biggest Problem in Getting Flood Information" — were Late or Delayed Information (26%) and Lack of Information or No Warnings at All (20%), together accounting for 38% of all responses. Respondents also cited unclear or inaccurate information (12%) and lack of location-specific flood susceptibility information (6%) as recurring barriers. The survey findings were subsequently validated through a Key Informant Interview with a staff member of the Bacoor Disaster Risk Reduction and Management Office (BDRRMO), who confirmed that the office's current flood information dissemination relies primarily on social media, email, and barangay coordination — and

acknowledged that resident engagement with official warning channels remains low due to poverty, work constraints, and low attendance at flood orientations.

The BDRRMO interview further revealed that while the office maintains historical flood data and GIS-based exposure maps per barangay, these datasets are not accessible to the general public in a usable or interactive format. The interviewee confirmed that the office's data holdings include barangay boundary maps, historical flood records (affected areas and flood events), and field-assessed exposure data — and expressed openness to data sharing for research and academic purposes. Critically, the BDRRMO officer affirmed that a mobile-based tool providing location-specific flood susceptibility information would serve a genuine community need — particularly for residents in highly susceptible barangays who require actionable flood awareness support for area-avoidance and evacuation decisions.

A supplementary problem-identification interview conducted on May 18, 2026 with the Officer-in-Charge of the Bacoar City Planning and Development Office (CPDCO) confirmed that the office holds land-use maps, elevation maps, flood-hazard maps, and barangay boundary maps, and that these can be released to the research team as hard-copy prints using the existing endorsement letter. The CPDCO officer further confirmed that Bacoar functions as a regional catch basin and low-lying area and that drainage degradation is a flood driver, validating the environmental features selected for the susceptibility model.

FloodSense is a proposed mobile application that addresses these identified gaps through the integration of machine learning-based flood susceptibility classification and augmented reality visualization. Using a supervised ML classification model trained on environmental features — including historical flood records, elevation data, and rainfall parameters — the system classifies flood susceptibility across Bacoar City's barangays into four levels: Low, Moderate, High, and Very High. These classifications are rendered through an augmented reality ground-plane overlay, allowing users to visualize susceptibility levels by pointing their phone camera at any flat surface.

Following the panel's challenge — "What happens next after users become aware?" — four additional action features were designed and incorporated: Check My Area, which uses GPS to detect the user's barangay and return a plain-language safety

status; an Action Checklist that provides level-specific preparedness guidance; Find Nearest Evacuation Center, which ranks BDRRMO-designated centers by distance and links to Google Maps; and Share Susceptibility Report, which generates a unique shareable URL for any barangay's susceptibility report accessible in any mobile browser. Together, these features transition FloodSense from a passive awareness tool to an active, actionable flood guidance system.

KEY FEATURES

Core Features (Existing)

1. Augmented Reality Flood Susceptibility Map

Renders a color-coded flood susceptibility map (Low/Moderate/High/Very High) per barangay through the phone camera using Vuforia ground-plane markerless tracking. The AR overlay is anchored to a detected flat surface — not GPS-linked to the user's outdoor position. A surface detection indicator (green = valid, red = invalid) ensures accurate rendering before the map is displayed.

- Color-coded susceptibility levels: Low (green), Moderate (yellow), High (orange), Very High (red)
- Markerless ground-plane tracking via Vuforia — works on any flat horizontal surface (table, floor)
- Surface detection indicator — green/red; map inputs locked until surface confirmed valid
- Pinch-to-zoom to examine individual barangay labels and susceptibility colors
- Legend visible in AR view; full metadata (STANDL elements) in info panel
- Disclaimer accessible from AR view: susceptibility levels indicate physical flood likelihood, not a real-time guarantee

2. PAGASA-Aligned Rainfall Input

Users select rainfall scenario and duration from dropdown menus using plain-language labels tied to PAGASA's public rainfall advisory classification. The app internally maps these selections to millimeter-per-hour values used by the ML model,

removing the need for technical numeric input from non-expert users. This design was directly validated by the BDRRMO key informant.

- Rainfall intensity presets: Light (<2.5 mm/hr), Moderate (2.5–7.5 mm/hr), Heavy (7.5–15 mm/hr), Intense (15–30 mm/hr), Torrential (>30 mm/hr)
- Duration dropdown: 1 hr, 3 hrs, 6 hrs, 12 hrs, 24 hrs
- Saved rainfall preferences auto-filled from previous session
- Reference: PAGASA Rainfall Advisory System

3. Machine Learning Flood Susceptibility Classification

A supervised classification model trained on environmental datasets classifies flood susceptibility per barangay based on user-selected rainfall parameters combined with pre-loaded spatial features. The model outputs one of four susceptibility levels per barangay zone, which directly drives the AR map color coding. Results are pre-computed and stored as static JSON lookup tables in Firestore — no live model inference occurs at runtime.

- Classification output: Low / Moderate / High / Very High
- Features: elevation, historical flood records, rainfall intensity and duration
- Data sources: BDRRMO historical flood records, CPDCO hard-copy maps (elevation, land use, hazard) supplemented by open digital elevation models, and PAGASA rainfall classification tables
- Model: Random Forest or XGBoost (scikit-learn, Python)

4. Results Summary Screen

A scrollable list of all Bacoor City barangays with color-coded susceptibility badges, populated from the pre-computed Firestore lookup table. Provides an at-a-glance overview without requiring the AR interface.

5. User Account and Preference System

Users create an account to access personalized features including saved rainfall preferences. Firebase Authentication handles user management with secure credential storage.

- Username and password account creation and login
- Home barangay registered at account creation

- Saved rainfall preference parameters (intensity and duration)

6. Onboarding and Disclaimer Screen

A first-launch onboarding screen informs users of the app's purpose, limitations, and proper usage — ensuring informed use and reducing the risk of misinterpreting susceptibility outputs as real-time flood guarantees. Re-accessible from the main dashboard at any time.

- App purpose and scope explanation
- Disclaimer: “This app models rainfall-induced flood susceptibility. It is not a real-time flood sensor. Always follow official warnings from BDRRMO and PAGASA.”
- Flat surface instruction for AR usage

New Action Features (Post-Panel Feedback)

In response to the defense panel's challenge — “What happens next after users become aware?” — four additional features were designed to transition FloodSense from a passive awareness tool to an active, actionable flood guidance system. All four operate within the existing Firebase free tier and require no additional paid APIs or services.

7. Check My Area — Location-Based Safety Status

After viewing the AR map or Results Summary, the user taps ‘Check My Area’. The app requests a one-time GPS fix and cross-references the user's coordinates against a static GeoJSON barangay boundary file to identify the user's current barangay. The pre-computed Firestore susceptibility lookup is then queried for that barangay under the currently selected rainfall scenario. A plain-language safety status card is displayed with the barangay name, susceptibility level color indicator, and a brief contextual message.

- GPS coordinates are used only for local on-device boundary lookup and are not stored or transmitted
- Compliant with Republic Act 10173 (Data Privacy Act)

- Example output (Low): ‘Your area (Brgy. Zapote) is LOW susceptibility under Heavy rainfall for 3 hours. Stay alert and monitor updates.’
- Example output (High): ‘Your area (Brgy. Talaba) is HIGH susceptibility under Heavy rainfall for 3 hours. Consider taking precautionary action.’

8. Action Checklist — What Should I Do?

Automatically displayed after the Safety Status card. Provides a tailored, level-specific set of concrete pre-event actions the user should take. Content is static and stored in the app or Firestore, mapped to each susceptibility level.

- LOW: Monitor weather updates; prepare go-bag; check drainage around property
- MODERATE: Secure important documents and valuables; identify nearest evacuation center; prepare go-bag; inform family of location and plan
- HIGH: Proceed to evacuation center if advised; avoid flood-prone roads and low-lying areas; bring go-bag and documents; alert relatives and neighbors; follow BDRRMO and PAGASA advisories
- VERY HIGH: Evacuate immediately; do not cross flooded roads or waterways; bring go-bag, documents, medicine, food, and water; contact relatives; follow BDRRMO and PAGASA emergency directives without delay

9. Find Nearest Evacuation Center

Triggered automatically when the user’s barangay is classified as Moderate, High, or Very High susceptibility. A ranked list of BDRRMO-designated evacuation centers is displayed, sorted by straight-line distance from the user’s GPS location using the Haversine formula computed on-device. Each entry shows the center name, barangay, and estimated distance. The user taps ‘Get Directions’ to be deep-linked into Google Maps for turn-by-turn navigation.

- Evacuation center list sourced from BDRRMO as part of the existing data-sharing collaboration
- No routing API or paid geolocation service required
- All navigation handled by Google Maps via standard geo: URI deep-link
- Same one-time GPS fix from Check My Area is reused — no additional location request

10. Share Susceptibility Report

Allows users to generate and share a full susceptibility report for any barangay in Bacoor City — including their own or a friend's or relative's location — via any messaging or social platform. The report is accessible as a mobile-friendly web page that anyone can open in a browser without installing the app.

- User selects target barangay from dropdown; rainfall parameters are pre-filled from the current session but can be adjusted
- App writes a report document to Firebase Firestore with a unique auto-generated ID
- A shareable URL is generated: `floodsense.web.app/report/{uniqueID}`
- URL is passed to Android's native share sheet — user sends via Viber, WhatsApp, SMS, Messenger, email, or any installed app
- Recipient opens the link in any mobile browser and views the full web-based report (barangay name, susceptibility level, action checklist, nearest evacuation center, disclaimer)
- Firebase Hosting serves a single static HTML/CSS/JavaScript file — no framework required
- Optional: Firestore documents set to auto-delete after 72 hours for storage hygiene

I. TECHNICAL FEASIBILITY

This section evaluates whether FloodSense can be built using currently available technologies, tools, and frameworks within the capability of an undergraduate Computer Science team. Each component is mapped to a specific, free or low-cost technology with a documented implementation path. The assessment covers both the original core features and the four new action features added in response to panel feedback.

1.1 Proposed Technology Stack

Layer	Technology	Cost	Justification
Mobile App (All Screens)	Unity + uGUI (Canvas, Input Fields, Dropdowns)	Free (Personal License)	Unified development environment for both UI screens and AR; eliminates Flutter/React Native handoff
AR Engine	Vuforia SDK (Ground Plane)	Free tier	Markerless ground-plane tracking on flat surfaces; no GPS anchoring required
Backend Database /	Firebase (Firestore, Auth, Hosting)	Free tier	Real-time sync, built-in authentication, cloud storage, and free hosting for the shareable web report page
ML Model	Python (pandas, NumPy, scikit-learn, XGBoost)	Free	Flood susceptibility classification; trained on open and BDRRMO/CPDCO datasets
ML Output Storage	Firebase Firestore (pre-computed JSON lookups)	Free	The trained model generates susceptibility lookup tables (JSON) for all supported rainfall intensity/duration combinations. These static files are stored in Firestore. The Unity app fetches the appropriate document directly
GPS Boundary Detection &	Android Location API + GeoJSON barangay boundary file	Free	One-time GPS fix used for Check My Area and Find Nearest Evacuation Center features. Boundary detection is performed locally on-device using a static GeoJSON file — no paid geolocation API required
Distance Calculation	Haversine Formula (on-device)	Free	Calculates straight-line distance from user's GPS position to each evacuation center stored in Firestore. No routing API required

Layer	Technology	Cost	Justification
Navigation (Directions)	Google Maps URI Deep-Link	Free	Passes destination coordinates to the installed Google Maps app. All routing is handled by Google Maps — no custom routing engine is built
Shareable Web Report	Firebase Hosting + Single HTML/CSS/JS file	Free tier	Static web page reads a Firestore document by unique ID and renders the full susceptibility report for the recipient. No framework required
Elevation / Spatial Data	CPDCO (via data-sharing agreement)	Free (institutional)	Topographic and land-use features for ML model training
Historical Flood Data	BDRRMO (via data-sharing agreement)	Free (institutional)	Ground-truth flood occurrence data per barangay; evacuation center list for Find Nearest Evacuation Center
Rainfall Reference	PAGASA Advisory Classification	Free (public)	Basis for user-facing rainfall intensity presets
UI/UX Prototyping	Figma	Free tier	Mobile interface design and prototyping prior to Unity development

1.2 Feature-Level Technical Assessment

Feature	Implementation Approach	Complexity	Est. Time
AR Flood Susceptibility Map (Vuforia Ground Plane)	Unity + Vuforia ground-plane detection; flat-surface anchored barangay zone overlay; color-coded susceptibility rendering	High	4 weeks
PAGASA Rainfall Dropdown + Susceptibility Lookup	Unity uGUI Dropdown mapped to mm/hr values; app reads the matching pre-computed susceptibility JSON document from Firestore (keyed by intensity + duration)	Medium	2 weeks
ML Classification Model (Python)	scikit-learn/XGBoost trained on elevation, historical flood records, and rainfall features. Pre-computed susceptibility lookup tables (JSON) generated for all supported rainfall intensity/duration combinations and uploaded to Firestore. No live model inference at runtime.	High	4 weeks
Results Summary Screen	Unity uGUI ScrollView / List of barangay names + color badge; fed directly from ML lookup stored in	Low	1 week

Feature	Implementation Approach	Complexity	Est. Time
	Firestore		
User Account + Auth (Firebase)	Firebase Authentication for email/password login; Firestore for user preferences and saved barangay	Low	1 week
Surface Detection Indicator	Vuforia ground-plane status callback; tint material green/red based on surface validity; lock input until green	Low	1 week
Onboarding + Disclaimer Screen	Unity first-launch scene with instructions and disclaimer; dismissable with user acknowledgement; re-accessible from main dashboard	Low	0.5 week
Check My Area (NEW)	Android Location API for one-time GPS fix; on-device GeoJSON boundary lookup to detect user's barangay; Firestore lookup for susceptibility level; plain-language safety status card rendered in Unity uGUI	Low-Medium	1 week
Action Checklist — What Should I Do? (NEW)	Static content screen in Unity uGUI; four preset text sets mapped to Low, Moderate, High, and Very High susceptibility levels. Triggered automatically after Check My Area result.	Very Low	2-3 days
Find Nearest Evacuation Center (NEW)	Static evacuation center JSON from BDRRMO uploaded to Firestore; Haversine formula for on-device distance calculation; ranked list rendered in Unity uGUI; 'Get Directions' button deep-links to Google Maps via geo: URI	Low-Medium	1 week
Share Susceptibility Report (NEW)	Firebase Firestore write for report document; unique URL generation (floodsense.web.app/report/{ID}); Android native share sheet integration; Firebase Hosting for single HTML web report page that reads Firestore by ID and renders full report	Medium	1.5-2 weeks
Flood Information Hub (stretch)	Static Firestore-backed content screen; barangay flood history, evacuation info, BDRRMO contacts	Low	1 week
Offline Mode	Locally cached ML model and	Medium	2 weeks

Feature	Implementation Approach	Complexity	Est. Time
(stretch)	barangay data; Firebase offline persistence for user data		

1.3 Technical Challenges and Mitigations

AR Rendering Performance on Mid-Range Devices

Vuforia ground-plane AR and barangay zone overlay rendering may cause performance issues on entry-level Android devices.

Mitigation: Optimize barangay zone geometry by simplifying polygon vertices. Implement level-of-detail rendering. Set minimum device specification in app description.

User Input Accuracy for Rainfall Selection

Users may not correctly identify rainfall intensity without reference, leading to inaccurate model queries.

Mitigation: Use PAGASA-aligned plain-language labels (Light, Moderate, Heavy, Intense, and Torrential) with optional rain intensity icons. Add brief instruction: 'Select the rainfall level that best matches current or expected conditions.' Directly validated by the BDRRMO key informant.

Flat Surface Detection Edge Cases

Users may point the camera at non-flat surfaces (grass, fabric, uneven floors), preventing ground-plane detection.

Mitigation: Vuforia's Ground Plane automatically detects surface validity. Implement green/red surface indicator and a persistent instruction message: 'Point camera at a flat table or floor.' Block input form from appearing until surface is confirmed green.

GPS Availability and Accuracy for Location-Based Features

GPS signals may be weak or unavailable indoors, affecting the accuracy of the Check My Area and Find Nearest Evacuation Center features.

Mitigation: Display a clear in-app message prompting users to move outdoors or to an area with better signal if GPS acquisition fails. The app performs a single one-

time GPS fix rather than continuous location tracking, minimizing battery drain and reducing the frequency of acquisition failures. Safety Status and Evacuation Center features can still be accessed manually through the Results Summary if GPS is unavailable.

GeoJSON Boundary Lookup Accuracy

Barangay boundary GeoJSON files derived from hard-copy maps may contain minor digitization inaccuracies, potentially misidentifying the user's barangay near boundary edges.

Mitigation: Cross-validate the GeoJSON boundary file against NAMRIA and CPDCO reference data. Inform users via the onboarding screen that boundary detection is approximate and that they should confirm their barangay manually if the detected result appears incorrect.

Shareable Report Link Accessibility

Recipients of a shared report link must have an internet connection to view the Firebase-hosted web report page.

Mitigation: The web report page is a lightweight single HTML file requiring minimal data. The disclaimer on the report page clearly states that the report is a pre-event modeled estimate and that users should follow official BDRRMO and PAGASA advisories.

II. OPERATIONAL FEASIBILITY

This section evaluates whether FloodSense can be realistically developed, deployed, and used within the constraints of the thesis timeline, team size, target users, and available institutional resources. The assessment reflects the revised feature set incorporating the four new action features added in response to panel feedback.

2.1 Team Composition and Role Distribution

Member	Primary Role	Responsibilities
Member 1	Mobile / AR Developer	Unity uGUI screen development (Login, Preferences, Results Summary, AR View, Check My Area, Action Checklist, Find Nearest Evacuation Center, Share Report screens); Unity scene navigation; user account UI; onboarding flow; Vuforia AR integration; surface detection UI; Android GPS and share sheet integration
Member 2	ML Engineer / Backend	Python ML model development (feature engineering, training, evaluation); Firebase setup; ML pre-computed lookup table generation and Firestore upload; Firebase Hosting setup; web report HTML page development; Firestore integration for new features
Member 3	Data / Research Lead	Spatial data collection and preprocessing (CPDCO, BDRRMO); barangay zone GeoJSON preparation; evacuation center data encoding; usability testing coordination; thesis documentation

2.2 Recommended Development Timeline (10 Months)

Phase / Period	Deliverables
Phase 1 — Foundation (Months 1–2)	Project setup; Firebase configuration; user authentication (email/password with username workaround); basic Unity UI screens (Login, Preferences placeholder, Results Summary placeholder); formal data request letters to BDRRMO and CPDCO; initial data preprocessing (cleaning, feature engineering); GeoJSON barangay boundary file acquisition or preparation
Phase 2 — ML Model Development (Months 2–4)	Data cleaning and feature engineering using pandas and NumPy (elevation, historical flood records, rainfall parameters); training and evaluation of Random Forest / XGBoost classifier (scikit-learn); generation of pre-computed susceptibility lookup tables (JSON) for all supported rainfall intensity/duration combinations; upload of JSON files to Firestore; encoding of BDRRMO evacuation center list into Firestore
Phase 3 —	Vuforia Unity ground-plane AR setup; flat-surface barangay zone overlay;

Phase / Period	Deliverables
AR Integration (Months 4–6)	color-coded susceptibility rendering; surface detection indicator; legend overlay; pinch-to-zoom interaction
Phase 4 — New Features Integration (Months 6–8)	Check My Area: Android GPS integration, GeoJSON boundary lookup, safety status card UI; Action Checklist: level-mapped static content screen; Find Nearest Evacuation Center: Haversine distance calculation, ranked list UI, Google Maps deep-link; Share Report: Firestore document write, unique URL generation, Android share sheet integration, Firebase Hosting web report page
Phase 5 — Testing & Refinement (Months 8–9)	Usability testing with at least 20 residents from the survey pool; BDRRMO review of susceptibility outputs and evacuation center data accuracy; bug fixes; ML model evaluation metrics write-up; final feature refinements
Phase 6 — Final Defense Preparation (Month 10)	Final defense preparation; documentation completion; thesis manuscript finalization

2.3 Target Users and Adoption Feasibility

FloodSense targets all residents of Bacoar City, with priority distribution through BDRRMO community outreach channels and direct recruitment from the original survey respondent pool.

- Tier 1 — Bacoar City Residents (flood-prone barangays): Residents of historically susceptible barangays (Zapote, Mambog, Talaba, Niog) who experience flooding regularly and need location-specific susceptibility information for evacuation and safety decisions. Pilot distribution through BDRRMO community outreach and barangay-level orientation; direct recruitment from survey respondents.
- Tier 2 — General Bacoar Residents: Residents outside the highest-susceptibility zones who use the app for general flood awareness and community preparedness planning. App store distribution; awareness through city-level flood information campaigns.

2.4 Financial Feasibility

FloodSense is designed to operate at zero licensing cost during the thesis development and pilot testing phases. The addition of four new action features introduces no new paid services — all new features operate within the existing Firebase free tier and rely on free platform APIs (Android Location API, Google Maps URI deep-link, GeoJSON boundary file).

Tool / Service	Purpose	Cost
Unity + Vuforia SDK	AR development, ground-plane tracking, all UI screens (uGUI)	Free (Personal License)
Firebase Spark Plan	Auth, Firestore database, Cloud Functions, Firebase Hosting	Free tier
Python / scikit-learn / XGBoost	ML model training and evaluation	Free
Android Location API	One-time GPS fix for Check My Area and Evacuation Center features	Free
Google Maps URI Deep-Link	Turn-by-turn navigation directions from Find Nearest Evacuation Center	Free
CPDCO Topographic Data	Elevation features for ML model	Free (data-sharing agreement)
BDRMO Historical Flood Data	Ground-truth barangay flood records; evacuation center list	Free (data-sharing agreement)
PAGASA Rainfall Advisory Data	Rainfall intensity classification reference	Free (public)
Figma	UI/UX prototyping	Free tier

2.5 New Features Feasibility Summary

The following table summarizes the feasibility of the four new features added in response to panel feedback. All four new features leverage the existing data backbone (Firestore lookup tables, Firebase Hosting) and require no additional paid APIs or services.

Feature	Complexity	Est. Dev Time	Key Dependency
Check My Area (Safety Status)	Low-Medium	1 week	GeoJSON barangay boundary file + existing Firestore lookup
Action Checklist —	Very Low	2–3 days	Static content only; no new API or

Feature	Complexity	Est. Dev Time	Key Dependency
What Should I Do?			data source required
Find Nearest Evacuation Center	Low-Medium	1 week	Evacuation center list from BDRRMO + Google Maps deep-link
Share Susceptibility Report	Medium	1.5–2 weeks	Firebase Hosting + single HTML web report page

2.6 Operational Risks and Mitigations

BDRRMO Data Access Delay

BDRRMO may require additional time to process a formal data-sharing request, potentially delaying ML model development and the encoding of evacuation center data required for the Find Nearest Evacuation Center feature.

Mitigation: Initiate formal data request immediately after title defense approval. Begin ML model development in parallel using PAGASA open datasets as primary sources. Treat BDRRMO data as supplementary validation input to avoid single-source dependency. Placeholder evacuation center data from publicly available BDRRMO communications can be used for development purposes, with the official list substituted upon data release.

CPDCO Spatial Data Inaccessibility

CPDCO may not have digital GIS files and shapefiles readily available, or may store data in formats incompatible with the research pipeline.

Mitigation: Hard-copy maps can be georeferenced and digitised using QGIS, converting them into usable vector or raster layers. If digitisation time exceeds the project schedule, open digital elevation models (SRTM, ALOS PALSAR) and NAMRIA land-cover data will be used directly.

Low Pilot User Adoption

Community residents, particularly elderly or low-literacy users, may find the AR interface or the new GPS-based features unfamiliar or difficult to use.

Mitigation: Conduct usability testing with at least 20 residents from the original survey pool. Simplify onboarding with visual instructions. The new action features

(Check My Area, Action Checklist, Evacuation Center) are designed as simple tap-based interactions following the AR map view, minimizing the learning curve. Leverage BDRMO community outreach as a distribution channel for pilot recruitment.

Scope Creep — Expanded Feature Set

The addition of four new features expands the development scope. Managing AR, ML, GPS integration, a web report page, and Android share sheet functionality in parallel is technically demanding for a thesis team.

Mitigation: The phased timeline separates ML completion (Phase 2), AR integration (Phase 3), and new feature development (Phase 4) to prevent bottlenecks. New features are architecturally lightweight: Check My Area and Action Checklist share the existing data backbone; Find Nearest Evacuation Center requires only a static JSON dataset and a Haversine formula; Share Report requires only a Firestore write and a single HTML file. Offline Mode and Flood Information Hub remain stretch goals deferred to Phase 4 only if earlier phases are completed ahead of schedule.

Data Privacy Compliance

User account data and GPS coordinates are subject to Philippine Republic Act 10173 (Data Privacy Act).

Mitigation: Firebase Security Rules enforce per-user data isolation. Informed consent obtained from all pilot participants. GPS coordinates used for Check My Area and Find Nearest Evacuation Center are processed on-device for local boundary lookup only and are not stored or transmitted to any external server. No personal location data is retained by the application.

Sustainability Post-Thesis

Without a formal data-sharing agreement, the app's ML model may become outdated as new flood events occur, and the evacuation center list may require periodic updates.

Mitigation: Frame sustainability as future work in the thesis defense. Proposed sustainability pathway: establish a formal MOU with BDRMO for periodic model retraining using updated historical flood records, and explore integration with

BDRRMO's existing community information dissemination channels for sustained distribution and evacuation center data maintenance.

Interview-Identified Gap: User Input Accuracy

As raised by the BDRRMO key informant, users cannot accurately estimate rainfall in millimeters, making free-text or numeric input unreliable.

Mitigation: Fully addressed through the PAGASA-aligned dropdown design — users select plain-language intensity levels (Light, Moderate, Heavy, Intense, and Torrential) tied to standardized mm/hr ranges. The app handles all technical conversion internally. This design decision was directly validated by the BDRRMO interview.

Interview-Identified Gap: Output Clarity for Non-Expert Users

The BDRRMO interviewee raised that users unfamiliar with map conventions may misread susceptibility levels as absolute flood predictions rather than classification-based outputs.

Mitigation: Include a brief legend explanation accessible from the AR view. Add disclaimer on first launch: 'Susceptibility levels indicate the likelihood of flooding based on environmental conditions — not a guarantee of actual flooding.' Brief instructional tooltip accessible during AR use. The plain-language safety status cards in Check My Area reinforce this framing with contextual messaging.

2.7 Proposed Future Enhancement: Conversational AI Assistant

During the design phase of the "What Should I Do?" Action Checklist feature, the potential for a conversational AI assistant was identified as a viable direction for future development. A context-aware AI chatbot integrated into the app could allow users to ask open-ended preparedness questions — such as "What should I bring if I need to evacuate with a toddler?" or "Is it safe to travel from Zapote to Molino right now?" — and receive personalized, dynamically generated responses informed by the user's current barangay, susceptibility level, and selected rainfall scenario.

This enhancement is technically achievable using the Anthropic Claude API or a comparable large language model API, with Firebase Functions serving as a

lightweight secure middleware layer to handle API calls without exposing credentials in the app. Context injection would allow the model to generate responses directly relevant to the user's flood situation rather than providing generic preparedness guidance.

However, this feature is deferred to future work for the following reasons: (1) LLM APIs operate on a pay-per-use pricing model, introducing ongoing costs beyond the zero-cost thesis development phase; (2) the feature introduces internet dependency for every AI query, which may affect usability in low-connectivity situations common during flood events; and (3) the conversational AI feature represents a meaningful scope addition that would extend the development timeline beyond what is feasible within the current 10-month window. Its inclusion as a future enhancement allows the research team to acknowledge and credit the concept without overextending the core thesis deliverables.

2.8 Feasibility Summary

Dimension	Assessment	Verdict
Technical	All features implementable with free, well-documented tools; ML susceptibility classification model trained on open and BDRRMO/CPDCO datasets using scikit-learn/XGBoost; AR rendering via Vuforia ground-plane tracking on flat surfaces — no GPS anchoring required; all screens built in Unity uGUI; new action features leverage existing Firestore data backbone with minimal additional complexity	FEASIBLE
Financial	Zero licensing cost during thesis period; all tools and frameworks operate within free tiers sufficient for thesis-scale pilot testing (20–50 participants); new features add no new paid APIs or services	FEASIBLE
Team Capacity	Three-person team with clearly separated AR/UI, ML/Backend, and Data roles; 10-month phased timeline isolates ML completion, AR integration, and new feature development into distinct phases to prevent bottlenecks	FEASIBLE
User Adoption	Pilot testing achievable through BDRRMO community outreach and direct recruitment from survey respondents; PAGASA-aligned plain-language input removes technical barrier; new action features (Check My Area, Action Checklist, Evacuation Center, Share Report) directly address the panel's concern that awareness alone is insufficient	FEASIBLE
Institutional Support	BDRRMO key informant confirmed openness to data sharing and collaboration; CPDCO confirmed on May 18, 2026 that required spatial maps exist and can be released as hard-copy prints; BDRRMO evacuation center data required for the Find Nearest Evacuation Center feature is obtainable under the existing data-sharing collaboration	FEASIBLE
Scope	Four new action features added in response to panel feedback, transitioning the system from passive awareness to active guidance; combined estimated development time for all new features is approximately 4–5 weeks, well within the 8-month development window; Flood Information Hub and Offline Mode remain stretch goals; conversational AI assistant deferred to future work	FEASIBLE
Academic Contribution	Novel integration of AR visualization and ML-based flood susceptibility classification for a Philippine LGU context; PAGASA-aligned community input design; BDRRMO-validated problem and concept; post-awareness action guidance system extending the study beyond susceptibility mapping into actionable community-level decision support	STRONG

